

Wave Insight™

AC Waveform Analyzer

Model: Wi-100



Quick Start Guide

English



AC Waveform Analyzer - Quick Start Guide

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1. Overview

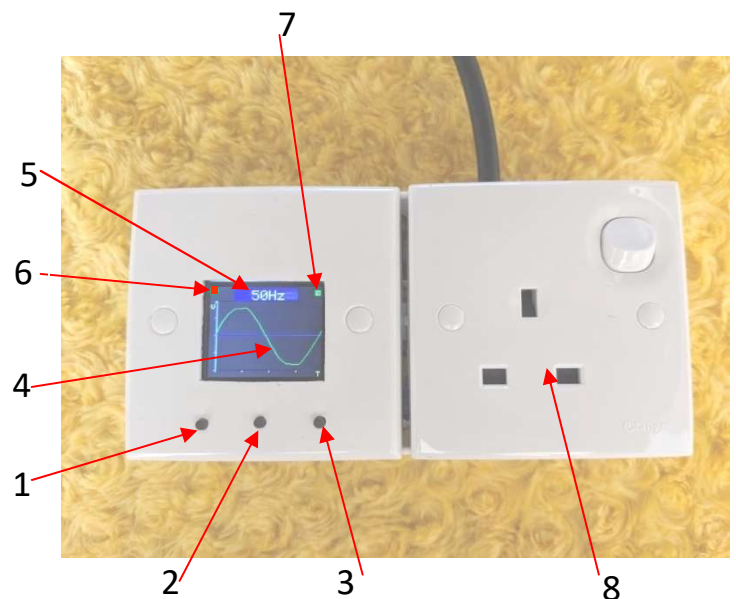
The AC Waveform Analyzer is a compact, microcontroller-based device designed to capture, display, and analyze AC voltage waveforms. It features a 1.8-inch TFT color display, three-button navigation, buzzer alert, and flash memory for saving user settings.

Key Features:

- Real-time waveform display
- Frequency measurement
- Reference waveform comparison
- Distortion detection with buzzer alert
- Adjustable brightness
- Adjustable sensitivity/tolerance
- Settings saved to internal flash memory

2. Device Layout

1. MENU button
2. DOWN button
3. UP button
4. Live AC Wave
5. Frequency
6. Reference save status icon
7. Buzzer status icon
8. AC power outlet (max 13A / 3000W)



3. Power On & Boot Sequence

When the device is powered on, the following sequence appears:

1. **Brand Logo Screen** - "Wave Insight™" logo
2. **Boot Screen** - AC Waveform Analyzer title with contact information
3. **Main Screen** - Waveform display with frequency readout



1



2



3

Screen Elements:

Element	Location	Description
Frequency	Top center	Current measured frequency in Hz
Waveform Graph	Center	Real-time voltage waveform
Reference Icon	Top-left	Green = Saved, Red = Not saved
Buzzer Icon	Top-right	Green = Buzzer ON, Red = OFF, Blink red = wave distortion alert
Y-Axis Label	Left	"V" for Voltage
X-Axis Label	Bottom-right	"T" for Time

4. Main Display Screen



The main screen shows the live waveform and key indicators.

5. Settings Menu

Press the **MENU (SET)** button **3 second** to enter the settings menu.

Menu Options:

#	Option	Description		
1	Buzzer	Configure buzzer timer	4	Wave Tolerance Set waveform comparison sensitivity
2	Brightness	Adjust display brightness	5	Reset All Reset all settings to factory defaults
3	Save Ref. wave	Save current waveform as reference	6	About View device information
			7	Exit Return to main screen

Navigation:

- **UP/DOWN** - Move cursor
- **MENU (SET)** - Select option / Go back

6. Buzzer Settings

The buzzer alerts when a waveform differs from the saved reference.

Option	Behavior
OFF	Buzzer disabled
ON	Buzzer sounds continuously when distortion detected
5 sec	Buzzer sounds for 5 seconds
10 sec	Buzzer sounds for 10 seconds
30 sec	Buzzer sounds for 30 seconds
1 min	Buzzer sounds for 60 seconds



Note: Buzzer requires a saved reference waveform to work (see Section 8).

7. Brightness Settings

Adjust the TFT display brightness.

Level	Brightness
25%	Dim
50%	Medium
75%	Bright
100%	Maximum



The brightness bar provides a visual indicator of each level.

8. Save Reference Waveform

This feature allows you to save the currently displayed waveform as a reference for comparison.

How to use:

1. Ensure a clean, undistorted waveform is displayed
2. Go to Settings → **Save Ref. wave**
3. Press MENU to save
4. The screen will flash white to confirm
5. A **green dot** will appear at the top-left of the main screen

9. Wave Tolerance (Sensitivity)

Adjust how much waveform deviation triggers the buzzer alert.

Value	Sensitivity	Use Case
0.01	Very High	Detect tiny changes
0.02	High	Sensitive detection
0.03	Medium-High	Moderate changes
0.05	Medium	Normal use
0.08	Normal	Normal
0.15	Low	Larger deviations
0.20	Very Low	Major changes only
0.30	Minimal	Severe distortion only



10. Reset All Settings

Restores all settings to factory defaults.

Resets:

- Buzzer: ON (continuous)
- Brightness: 25%
- Sensitivity: 0.05
- Reference waveform: Cleared

11. About

Displays device information including:

- Product name
- Manufacturer
- Contact

12. Specifications

1. **Plug & Play Operation:** Simply connect directly to a 230V AC wall socket. No external batteries or power adapters are required.
2. **One-Touch Reference Comparison:** Save a known-good waveform with a single button press. If the live waveform deviates beyond the set tolerance, the device instantly alerts you.
3. **Pass-Through Power Socket:** Features a unique "Dual Gang" design. While the analyzer is plugged in and monitoring, the front-facing extra socket keeps your wall outlet free for other appliances.
4. **Compact & Truly Portable:** Its compact form factor fits easily into any standard toolbox or work bag, making it perfect for field technicians and on-site diagnostics.

13. Safety and Instructions

WARNING:

- Always **unplug** the device from the mains socket before check fuse. If fuse is bad, replace with same value (0.1A / 100mA)
- **HIGH VOLTAGE:** This device operates on 230V AC mains. Electric shock hazard exists if enclosure is damaged or opened.
- **Do NOT** open the enclosure, No User Serviceable Parts inside
- **Keep away from water, rain, and excessive moisture.**
- **DO NOT** operate with wet hands or in damp/wet environments.
- **Inspect the power cord regularly. If damaged, discontinue use immediately.**
- **Use only a properly grounded quality 230V AC main socket**
- **Don't remove any sticker, warranty void if remove**

14. Troubleshooting

Problem	Possible Cause	Solution
No display	Power issue	Check power connection & fuse
Display too dim	Low brightness	Increase brightness in Settings
Buzzer not working	No reference saved	Save a reference waveform first
False buzzer alerts	Sensitivity too high	Increase tolerance value
Screen freezes	Menu active	Press MENU to exit